G2C -- Generate-to-Class Framework

Agentic AI Generator Classes for E2A, A2C, and P0 Frameworks

E2A-governed generators that produce E2A abstract classes, A2C abstract classes, and inherited concrete implementations via LLM

Extends: A2C Framework | Requires: E2A Framework, P0 Framework | Generator runtime: Python

github.com/subhamviky/a2c-framework

Author: Subham Gupta, Staff Architect & AI Architect

1. EXECUTIVE SUMMARY

The G2C (Generate-to-Class) Framework is the top layer of the framework stack. It provides a set of agentic AI generator classes that use LLM calls to generate E2A abstract classes, A2C abstract classes, and their concrete inherited implementations. Every generator class is written in Python. Every generated output is runtime-agnostic -- Python, Java, Node, or Go -- determined entirely by the runtime parameter in the request.

A developer who needs a new E2A agent, an A2C SDLC workflow, or a complete scaffolded project with business logic provides a single JSON config or natural-language prompt. G2C resolves or generates the abstract class, scaffolds the project structure via P0, generates the business logic via LLM, validates the output through a GeneratorCriticAgent quality gate, and writes all files. One call. One result.

The Self-Referential Design Principle

All G2C generator classes are themselves E2A-governed agents. Every generator inherits from BaseGeneratorAgent, which follows the same Template Method Pattern as E2A's BaseAgent: one PUBLIC entry point (generate()), PROTECTED domain hooks, and PRIVATE infrastructure methods (__write_file, __ensure_idempotency) and a model call delegated to a BaseLLMAgentService. The agent is governed by E2A. The output follows E2A. The framework stack is self-generating.

Layer	Framework	Entry Point	What it produces	Built on
1	**E2A**	agent.run()	Governed agentic AI agents, RAG pipelines, tool services	--
2	**A2C**	sdlc_agent.run()	Microservice code, Terraform IaC, GitHub Actions CI/CD	E2A
3	**P0**	bootstrapper.bootstrap()	Project structure, build manifests, Dockerfile, Makefile	Alongside A2C
4	**G2C** (this document)	generator.generate()	E2A/A2C abstract classes, inherited class files, scaffolded projects	E2A + A2C + P0

2. DESIGN PRINCIPLES

2.1 Four Generator Classes

G2C provides four generator classes, each with a single responsibility:

Generator Class	generator_type	Generates	P0 scaffold?
E2AAbstractClassGenerator	'e2a_abstract'	e2a_base.py or E2ABase.java -- the E2A abstract class foundation	No -- standalone file only
A2CAbstractClassGenerator	'a2c_abstract'	a2c_base.py -- SDLCAssistantAgent, DevRequest, SDLCWorkflow	No -- standalone file only
E2AInheritedClassGenerator	'e2a_inherited'	Abstract class (if needed) + P0 project scaffold + inherited agent/workflow class	Yes
A2CInheritedClassGenerator	'a2c_inherited'	E2A + A2C abstract classes (if needed) + P0 scaffold + SDLCAssistantAgent subclass	Yes

2.2 BaseGeneratorAgent -- Shared Base

All four generators inherit from BaseGeneratorAgent(ABC). Cross-cutting concerns -- LLM delegation (to BaseLLMAgentService), idempotency, latency SLO, token budget, quality gate, file writing, and observability -- are enforced once in the base class via the Template Method Pattern. Subclasses override only domain-specific protected methods.

2.3 Runtime Agnosticism

The generator class is written in Python. The generated output is runtime-agnostic. The same E2AAbstractClassGenerator instance generates a Python abstract class or a Java abstract class based solely on request['runtime']. The generator never executes, compiles, or imports the code it generates. The _write_output() method writes the LLM response string to a file with the runtime-correct extension.

request['runtime']	Output file	Language pattern	Import style in generated code
'python'	e2a_base.py	Python 3.x ABC with @abstractmethod decorators	from common.e2a_base import BaseAgent
'java'	E2ABase.java	Java 21 abstract class with final template methods	import com.company.common.E2ABase.BaseAgent
'node'	e2aBase.ts	TypeScript abstract class with abstract methods	import { BaseAgent } from './e2aBase'
'go'	e2a_base.go	Go interface + struct with embedded base methods	import 'github.com/org/e2a'

2.4 Abstract Class Source -- Three Modes

For inherited class generators, the abstract class can come from three sources. If abstract_class_source is not provided, it defaults to 'generate' -- the abstract class is created automatically before the inherited class generation begins.

Source mode	When to use	abstract_class_source value
Generate (default)	No abstract class exists yet. Generator calls E2AAbstractClassGenerator internally and places the output in common/ before generating the inherited class.	'generate' (default when not specified)
Path	Abstract class file already exists locally. Generator reads from abstract_class_path and uses the content as LLM context.	'path'
Package	Abstract class published as pip/Maven package. Generator produces the inherited class with the correct package import statement.	'package' + package_name field

2.5 System Prompt Auto-Build

The system prompt encoding E2A NFR guardrails, governance, and reliability patterns is always auto-built from versioned prompt files stored in prompts/. Whether or not system_prompt_additions is provided by the developer, the full E2A framework specification is always present in the system prompt. system_prompt_additions only adds domain/team-specific rules on top of the mandatory base.

2.6 Abstract Class Distribution -- Standalone vs Library

Abstract classes can be distributed as standalone files or as published packages. G2C supports both distribution modes through the abstract_class_source field. Publishing as a library is a separate CI/CD concern outside the generator scope.

Aspect	Standalone file (current)	Library package (design supported)
abstract_class_source	'generate' or 'path'	'package'
Location	common/ directory in project repo	Installed pip/Maven dependency
Import style	from common.e2a_base import BaseAgent	from e2a_framework import BaseAgent
Multi-team sharing	Copy common/ file to each repo	All teams install same package version
G2C support	Fully implemented	Design supported via 'package' source mode; publishing out of scope now

2.7 Quality Gate on All Paths

Every generator's generate() sequence ends with a GeneratorCriticAgent call that scores the generated output 0.0-1.0. A score below 0.75 triggers _fallback() and regeneration with a stricter prompt. The critic validates structural correctness: access modifier rules, required abstract methods, correct inheritance, NFR patterns in code not comments.

3. TYPEDDICTS -- INPUT AND OUTPUT CONTRACTS

3.1 GeneratorRequest -- Unified Input

A single TypedDict accepted by all four generators. The generator_type field routes to the correct class via GeneratorFactory. All fields not explicitly provided resolve to safe defaults.

```
class GeneratorRequest(TypedDict, total=False):
    # -- Universal (all generators) ------------------------------------------
    generator_type:          str       # 'e2a_abstract' | 'a2c_abstract'
                                       # | 'e2a_inherited' | 'a2c_inherited'
    source:                  str       # 'json' | 'prompt' | 'inline'  (default: 'inline')
    config path:             str       # path to generator-config.json (if source='json')
    runtime:                 str       # 'python' | 'java' | 'node' | 'go'
    runtime_version:         str       # '3.12' | '21' | '20' | '1.22'
    idempotency:             bool      # include idempotency pattern  (default: True)
    async_mode:              bool      # async vs sync execution      (default: True)
    platform:                str       # 'aws' | 'gcp' | 'azure' | 'standalone'
    domain:                  str       # 'financial' | 'logistics' | 'e-commerce'
    team:                    str       # team name for governance scoping
    company:                 str       # company name for governance scoping
    output_path:             str       # base output directory

    # -- Prompt customisation ------------------------------------------------
    system_prompt_additions: str       # domain NFRs, governance rules to add to base prompt
    user prompt:             str       # developer natural-language requirement
    mandatory_nfrs:          List[str] # additional NFRs beyond E2A defaults
    optional_nfrs:           List[str]

    # -- Abstract-class-specific (E2AAbstractClassGenerator, A2CAbstractClassGenerator) --
    base_class_name:         str       # 'BaseSettlementAgent' | 'BaseOrderProcessor'
    abstract_methods:        List[str] # method names beyond E2A defaults
    class description:       str       # one-line description for generated docstring

    # -- Inherited-class-specific (E2AInheritedClassGenerator, A2CInheritedClassGenerator) --
    agent_name:              str       # 'RefundAgent' | 'SettlementOrchestrator'
    abstract_class_source:   str       # 'generate' (default) | 'path' | 'package'
    abstract_class_path:     str       # required if abstract_class_source='path'
    abstract class config:   dict      # if abstract class source='generate'
    package_name:            str       # if abstract_class_source='package'
    scaffold:                dict      # ScaffoldRequest from P0 Framework

    # -- LLM config ----------------------------------------------------------
    model_id:                str       # main model  (default: claude-3-5-sonnet)
    critic_model_id:         str       # quality gate (default: claude-3-haiku)

    # -- Written by generator (do not supply) --------------------------------
    generator_key:           str       # idempotency key: name:type:runtime:platform
    generated_files:         List[str] # absolute paths of all created files
    critic_report:           dict      # {score, missing, passed}
    errors:                  List[str]
```

3.2 GeneratorResult -- Output Contract

```
class GeneratorResult(TypedDict, total=False):
    success:              bool
    generator key:        str
    generated_files:      List[str]   # all created file paths
    abstract_class_path:  str         # path to abstract class (inherited generators)
    scaffold_result:      dict        # ScaffoldResult from P0 (if P0 was invoked)
    critic_report:        dict        # {score, missing, passed} from GeneratorCriticAgent
    trace_id:             str         # UUID correlation ID
    latency_ms:           float
    errors:               List[str]
```

3.3 Minimum Viable Request per Generator

Based on default behaviour rules: if abstract_class_source is not provided it defaults to 'generate'; if system_prompt_additions is not provided the full E2A base system prompt is used unchanged; if idempotency or async_mode are not provided they default to True.

Generator	Minimum required fields	What defaults automatically
E2AAbstractClassGenerator	generator_type, runtime, output_path	idempotency=True, async_mode=True, system prompt from versioned file
A2CAbstractClassGenerator	generator_type, runtime, output_path	Same as above + E2A abstract class resolved automatically
E2AInheritedClassGenerator	generator_type, runtime, agent_name, output_path	abstract_class_source='generate', P0 scaffold from defaults, system prompt from file
A2CInheritedClassGenerator	generator_type, runtime, agent_name, output_path	Same + E2A and A2C abstract classes generated if not provided

4. BASEGENERATORAGENT -- ABSTRACT BASE CLASS

4.1 Class Variables

Variable	Type	Default	Purpose
generator_name	str	'BaseGeneratorAgent'	Governance key -- set in every subclass, used in all log entries
idempotency	bool	True	Same generator_key = skip. Set False to regenerate an existing output.
APPROVED_RUNTIMES	List[str]	['python','java','node','go']	Validated against request['runtime'] at Step 3
APPROVED_GENERATOR_TYPES	List[str]	['e2a_abstract','a2c_abstract','e2a_inherited','a2c_inherited']	Governance allow-list
PROMPT_DIR	str	'prompts/'	Base directory for versioned system prompt files

4.2 Method Contract Table

ACCESS	METHOD / SIGNATURE	RETURNS	PURPOSE
PUBLIC	`generate(request, config, **kwargs)`	`GeneratorResult`	Single entry point. Fixed 12-step sequence. Never override in subclasses.
ABSTRACT	`_validate_request(request, config, **kwargs)`	`bool`	Check required fields for this generator type. Return False to abort.
ABSTRACT	`_resolve_config(request, config, **kwargs)`	`GeneratorRequest`	Load from JSON file or LLM prompt if source != inline.
ABSTRACT	`_build_system_prompt(request, config, **kwargs)`	`str`	Load versioned base prompt + merge system_prompt_additions.
ABSTRACT	`_build_user_prompt(request, config, **kwargs)`	`str`	Construct developer-facing user prompt from request fields.
PROTECTED	`_apply_policy(request, config, **kwargs)`	`None`	Governance: approved runtime, approved generator type, NFR allow-lists.
ABSTRACT	`_evaluate_output(response, request, config, **kwargs)`	`float`	Score generated output 0.0-1.0 via GeneratorCriticAgent.
PROTECTED	`_fallback(request, config, **kwargs)`	`str`	Stricter regeneration prompt when critic score < 0.75.
ABSTRACT	`_write_output(response, request, config, **kwargs)`	`List[str]`	Write generated files to output_path. Return list of file paths.
PROTECTED	`_handle_error(error, request, result, config, **kwargs)`	`GeneratorResult`	Log error, append to errors[], return safe result.
PROTECTED (base-owned)	`_infer(messages, config, **kwargs)`	`dict`	Delegates the model call to config['llm_service'] (a BaseLLMAgentService). No provider routing or SDK code in the generator.
PRIVATE	`__ensure_idempotency(request)`	`str`	Returns generator_key. Skip generate() if already generated.
PRIVATE	`__track_tokens(messages)`	`int`	FinOps token count for budget enforcement.
PRIVATE	`__write_file(root, path, content)`	`str`	Idempotent file writer. Returns absolute path.

ACCESS	METHOD / SIGNATURE	RETURNS	PURPOSE
PRIVATE	`__generate_key(request)`	`str`	Business idempotency key: name:type:runtime:platform
PRIVATE	`__load_prompt_file(filename)`	`str`	Load versioned system prompt file from PROMPT_DIR.

4.3 generate() -- 12-Step Fixed Sequence

Step	Action	Condition
1	Resolve config from source (JSON/prompt/inline)	If source != 'inline'
2	Validate GeneratorRequest via _validate_request()	Always -- False aborts with ValueError
3	Governance approval check (runtime, generator_type in approved lists)	Always
4	Idempotency check -- skip if already generated	Only if self.idempotency = True
5	Build system prompt via _build_system_prompt() + runtime pattern injection	Always
6	Build user prompt via _build_user_prompt()	Always
7	Token budget check via __track_tokens()	Always
8	LLM call via _infer(messages, config) -> config['llm_service']	Always
9	Quality gate via _evaluate_output() -> GeneratorCriticAgent	Always
10	Fallback regeneration via _fallback() if critic score < 0.75 (max 2 retries)	Only if score < 0.75
11	Write output files via _write_output()	Always on success
12	Latency SLO check + structured observability log (trace_id, latency, tokens, score)	Always

4.4 Runtime Pattern Injection

The _build_system_prompt() method injects runtime-specific code patterns into the versioned system prompt via a RUNTIME_PATTERNS dictionary. This ensures the generated output uses correct language idioms regardless of which generator subclass is active.

```
RUNTIME_PATTERNS = {
    'python': {
        'abstract_base':    'from abc import ABC, abstractmethod',
        'abstract_method':  '@abstractmethod',
        'private_prefix':   '__',
        'protected_prefix': '_',
        'type_hints':       'from typing import Any, Dict, List, Optional'
    },
    'java': {
        'abstract_base':    'public abstract class',
        'abstract_method':  'protected abstract',
        'private_prefix':   'private',
        'protected prefix': 'protected',
        'type_hints':       'import java.util.*'
    },
    'node': {
        'abstract_base':    'abstract class',
        'abstract_method':  'abstract',
        'private prefix':   '#',
        'protected_prefix': 'protected',
        'type_hints':       'import type { ... }'
    },
    'go': {
        'abstract_base':    'type BaseAgent interface',
        'abstract method':  '// interface method',
        'private_prefix':   'lowercase',
        'protected_prefix': 'lowercase',
```

```
        'type_hints':       ''
    }
}

# Usage in _build_system_prompt():
def _build_system_prompt(self, request, config=None, **kwargs) -> str:
    base = self.__load_prompt_file(f'{self.generator_name.lower()}_system_v1.txt')
    runtime = request.get('runtime', 'python')
    patterns = RUNTIME_PATTERNS.get(runtime, RUNTIME_PATTERNS['python'])
    additions = request.get('system_prompt_additions', '')  # empty = base prompt only
    return base.format(
        runtime=runtime,
        runtime_version=request.get('runtime_version', '3.12'),
        runtime_patterns=patterns,
        idempotency=request.get('idempotency', True),
        async_mode=request.get('async_mode', True),
        domain=request.get('domain', 'general'),
        system_prompt_additions=additions
    )
```

5. E2AABSTRACTCLASSGENERATOR

5.1 Purpose and Output

Generates the E2A base abstract class file -- the foundation layer of a new E2A application. P0 scaffolding is not performed. The output is a single standalone file designed to be placed in a project's common/ directory or published as a package.

Field	Value
generator_name	'E2AAbstractClassGenerator'
generator_type (request field)	'e2a_abstract'
Output -- Python	e2a_base.py containing BaseAgent, BaseWorkflow, BaseRAGPipeline, BaseToolService, BaseLLMAgentService, and four Foundation interfaces
Output -- Java	E2ABase.java with abstract classes and final template methods
System prompt file	prompts/e2aabstractclassgenerator_system_v1.txt
Critic validates	All 9 abstract classes present; Single Public Entry Point rule enforced; all access modifiers correct; NFR matrix complete in base class

5.2 System Prompt -- E2A Abstract Generator

The system prompt encodes the complete E2A framework specification. It is loaded from the versioned file and formatted with runtime-specific values. Even if system_prompt_additions is empty, the full E2A specification is always present.

```
# prompts/e2aabstractclassgenerator_system_v1.txt (structure -- full content in repo)

You are a Principal Cloud Architect generating production-grade {runtime} code.
Generate the E2A Architecture Framework base abstract class file.

MANDATORY STRUCTURE (Template Method Pattern):
  - Every class has exactly ONE public entry point per interaction mode: run() / execute() /
retrieve();   BaseLLMAgentService alone has two: process_inference() (unary) and
process_inference_stream() (streaming)
  - PROTECTED methods ({runtime_patterns[protected_prefix]}method): override in subclasses
  - PRIVATE methods ({runtime_patterns[private_prefix]}method): framework internals only
  - CLASS VARS: idempotency = {idempotency}, async_mode = {async_mode}

MANDATORY ABSTRACT CLASSES (generate all 9):
  BaseAgent           run()      agents/
  BaseWorkflow        execute()  orchestration/
  BaseRAGPipeline     retrieve() rag/
  BaseToolService     execute()  tools/
  BaseLLMAgentService  process_inference() + process_inference_stream()  llm/ (model call
lives here ONLY; agents delegate via config['llm_service'])
  BaseInfraProvisioner           foundation/ (Interface -- no shared implementation)
  BaseObservability              foundation/ (Interface)
  BasePipeline                   foundation/ (Interface)
  BaseGovernanceFramework        foundation/ (Interface)

MANDATORY NFRs (enforce structurally in base class, NOT as comments):
  Idempotency:   __ensure_idempotency() in run()/execute()/retrieve()
  Latency SLO:   time delta check, raise NFRViolationError if exceeded
  Token budget:  __track_tokens(), raise if > max_tokens
  Governance:    approved_agents/models/tools list check
  Observability: structured JSON log (trace_id, latency, tokens, confidence)
  Fallback:      _fallback() triggered when confidence < min_confidence
  Async/sync:    async mode class variable toggle in public method
  Capacity:      pod semaphore + RPM/TPM token bucket enforced in BaseLLMAgentService, never
in agents

DOMAIN: {domain}  |  RUNTIME: {runtime} {runtime_version}
ADDITIONAL NFRs: {system_prompt_additions}
```

5.3 Key Method Overrides

```
class E2AAbstractClassGenerator(BaseGeneratorAgent):
    generator name = 'E2AAbstractClassGenerator'

    def _validate_request(self, request, config=None, **kwargs) -> bool:
        return all(k in request for k in ['runtime', 'output_path'])

    def _resolve_config(self, request, config=None, **kwargs) -> GeneratorRequest:
        if request.get('source') == 'json':
            loaded = self._BaseGeneratorAgent__load_json_config(request['config_path'])
            request.update(loaded)
        return request

    def _evaluate_output(self, response, request, config=None, **kwargs) -> float:
        critic = GeneratorCriticAgent(config={'critic_type': 'e2a_abstract'})
        return critic.score(response, request)

    def _write_output(self, response, request, config=None, **kwargs) -> List[str]:
        filemap = {'python': 'e2a_base.py', 'java': 'E2ABase.java',
                   'node': 'e2aBase.ts', 'go': 'e2a_base.go'}
        fname = filemap.get(request.get('runtime','python'), 'e2a_base.py')
        path = self._BaseGeneratorAgent__write_file(
            request['output_path'], fname, response
        )
        return [path]

    # _build_system_prompt and _build_user_prompt use base class helpers
    # run() is NOT overridden -- inherits 12-step sequence from BaseGeneratorAgent
```

6. A2CABSTRACTCLASSGENERATOR

6.1 Purpose and Output

Generates the A2C base abstract class file -- SDLCAssistantAgent, DevRequest TypedDict, CodeCriticAgent, IaCAgent, CICDAgent, and the SDLCWorkflow LangGraph StateGraph. No P0 scaffolding performed. The E2A abstract class is required as context; if not provided, it is generated automatically.

Field	Value
generator_name	'A2CAbstractClassGenerator'
generator_type	'a2c_abstract'
Output -- Python	a2c_base.py containing SDLCAssistantAgent, DevRequest TypedDict, SDLCWorkflow, CodeCriticAgent
System prompt file	prompts/a2cabstractclassgenerator_system_v1.txt
Requires	E2A abstract class content as LLM context -- abstract_class_source applies
Critic validates	SDLCAssistantAgent present, DevRequest TypedDict correct, _apply_policy() injects mandatory NFRs, CodeCriticAgent scores >= 0.75, SDLCWorkflow chains all 5 agents

6.2 Key Difference -- Requires E2A Context

A2CAbstractClassGenerator always includes the full e2a_base.py content in the system prompt so the LLM generates A2C classes that correctly extend BaseAgent. The abstract_class_source field determines how e2a_base.py is obtained. If not provided, E2AAbstractClassGenerator is called first and the result is injected automatically.

```
def  build system prompt(self, request, config=None, **kwargs) -> str:
    base = self.__load_prompt_file('a2cabstractclassgenerator_system_v1.txt')
    # Load E2A abstract class content for LLM context
    e2a_content = self.__resolve_abstract_class_content(request, config)
    additions = request.get('system_prompt_additions', '')
    return base.format(
        runtime=request.get('runtime','python'),
        runtime_patterns=RUNTIME_PATTERNS.get(request.get('runtime','python')),
        e2a_abstract_class=e2a_content,   # full e2a_base.py injected
        idempotency=request.get('idempotency', True),
        async_mode=request.get('async_mode', True),
        system_prompt_additions=additions
    )

def __resolve_abstract_class_content(self, request, config) -> str:
    src = request.get('abstract_class_source', 'generate')  # default: generate
    if src == 'path':
        with open(request['abstract_class_path']) as f: return f.read()
    elif src == 'package':
        return f'# Package: {request["package_name"]} -- import from installed lib'
    else:  # 'generate'
        result = E2AAbstractClassGenerator(config=config).generate(
            {**request.get('abstract_class_config',{}),
             'generator_type':'e2a_abstract',
             'output path': request.get('output path','./') + 'common/'},
            config
        )
        with open(result['generated_files'][0]) as f: return f.read()
```

7. E2AINHERITEDCLASSGENERATOR

7.1 Purpose and Output

Generates a complete E2A inherited class project in a single call: resolves or generates the abstract class, runs P0 scaffolding to create the project structure, then generates the concrete inherited agent/workflow/RAG/tool class implementing the developer's business logic.

Field	Value
generator_name	'E2AInheritedClassGenerator'
generator_type	'e2a_inherited'
Output	Abstract class file in common/ + full P0 project scaffold + inherited class file(s)
System prompt file	prompts/e2ainheritedclassgenerator_system_v1.txt
Critic validates	Inherits from correct base, all required protected methods overridden, run() NOT overridden, NFRs in code not comments, no direct __private calls

7.2 Internal Execution Phases

Phase	Steps	What happens
Setup	1-4	Resolve config, validate, governance check, idempotency
Abstract class resolution	5-6	Resolve e2a_base.py via generate/path/package; place in project's common/ directory
P0 scaffold	7-8	Call ProjectBootstrapperFactory.create(request['scaffold']).bootstrap(); creates pyproject.toml/pom.xml, directories, Dockerfile, Makefile
Logic generation	9-11	Build system prompt (E2A rules + abstract class content + domain context); build user prompt; call _infer()
Quality gate	12	GeneratorCriticAgent validates inherited class structure; fallback if score < 0.75
Output	13-14	Write inherited class file into scaffolded project; latency SLO + observability log

7.3 Abstract Class Directory Placement

When the abstract class is generated (source='generate') or loaded from a path, it is placed in the project's common/ directory. The generated inherited class uses the correct import pattern for the runtime:

```
# Python -- abstract class at {project_root}/common/e2a_base.py
from common.e2a_base import BaseAgent, AgentState

class SettlementAgent(BaseAgent):
    agent_name = 'SettlementAgent'
    ...

# Java -- abstract class at {project_root}/common/E2ABase.java
import com.company.common.E2ABase.BaseAgent;

public class SettlementAgent extends BaseAgent {
    public static final String AGENT_NAME = "SettlementAgent";
    ...
}
```

7.4 System Prompt Design -- Inherited Class

```
# prompts/e2ainheritedclassgenerator_system_v1.txt (structure)
```

```
You are a Principal Cloud Architect generating a production-grade {runtime}
inherited class that extends the E2A abstract class below.

E2A FRAMEWORK RULES:
  - PROTECTED methods: override _validate_state, _build_messages, _evaluate_output
  - DO NOT override run()/execute()/retrieve() -- inherited from base class
  - DO NOT call any __private method from outside the class
  - _apply_policy() only if the agent has governance rules
  - _fallback(): safe degraded response without LLM call
  - _handle_error(): append to state['errors'], return safe state

ABSTRACT CLASS TO INHERIT FROM:
{abstract_class_content}

DOMAIN: {domain}  |  AGENT NAME: {agent name}
ADDITIONAL NFRs: {system_prompt_additions}
```

8. A2CINHERITEDCLASSGENERATOR

8.1 Purpose and Output

Generates a complete A2C inherited class project: resolves or generates both the E2A and A2C abstract classes, runs P0 scaffolding, then generates a concrete SDLCAssistantAgent subclass implementing the developer's custom SDLC workflow. Used when a team needs their own code generation agent with domain/team-specific NFR injection.

Field	Value
generator_name	'A2CInheritedClassGenerator'
generator_type	'a2c_inherited'
Output	E2A abstract class + A2C abstract class + P0 scaffold + SDLCAssistantAgent subclass
System prompt file	prompts/a2cinheritedclassgenerator_system_v1.txt
Requires	Both e2a_base and a2c_base as LLM context -- both resolved via abstract_class_source
Critic validates	Extends SDLCAssistantAgent, _apply_policy() injects team-specific NFRs, DevRequest fields correct, run() not overridden

8.2 Key Difference -- Requires Both E2A and A2C Context

The system prompt for A2CInheritedClassGenerator includes both e2a_base.py and a2c_base.py content. The LLM generates a class that extends SDLCAssistantAgent (not BaseAgent directly) and adds team/domain-specific _apply_policy() logic on top of A2C's mandatory NFR injection.

```
# Generated A2C inherited class (Python example):
from a2c_base import SDLCAssistantAgent

class FinancialSettlementSDLCAgent(SDLCAssistantAgent):
    agent name = 'FinancialSettlementSDLCAgent'

    def _apply_policy(self, state, config=None, **kwargs) -> None:
        # Call parent to inject A2C mandatory NFRs first
        super()._apply_policy(state, config, **kwargs)
        # Add team-specific NFRs on top
        team_nfrs = ['DoubleEntryLedgerInvariant', 'SagaCompensation', 'AuditTrail']
        for nfr in team_nfrs:
            if nfr not in state.get('mandatory_nfrs', []):
                state['mandatory nfrs'].append(nfr)

    def _build_messages(self, state, config=None, **kwargs) -> list:
        # Domain-specific generation context
        return [
            {'role':'system','content': f'Generate SAP TM financial settlement '
             f'microservice: {state.get("context","")}'},
            {'role':'user','content': f'Project: {state.get("project_name")}'}
        ]
```

9. GENERATORCRITICAGENT -- QUALITY GATE

9.1 Purpose

GeneratorCriticAgent validates the structural correctness of every generated class. It is called at Step 9 of every generator's sequence. A score below 0.75 triggers _fallback() and regeneration. The critic is itself an E2A-governed agent inheriting from BaseAgent, using a cheaper model (e.g. claude-3-haiku) for cost efficiency.

9.2 Critic Types and Validation Rules

critic_type	Validates	Scoring
'e2a_abstract'	All 9 abstract classes present; single public entry point per class; PROTECTED methods single-underscore; PRIVATE methods double-underscore; NFR enforcement in base class code, not comments; Foundation classes are interfaces with no shared implementation	Deduct 0.15 per missing class; 0.10 per access modifier violation
'a2c_abstract'	SDLCAssistantAgent present; DevRequest TypedDict fields correct; _apply_policy() injects mandatory NFRs; CodeCriticAgent integration present; SDLCWorkflow chains all 5 agents	Same deduction model
'e2a_inherited'	Inherits from correct base; all required protected methods overridden; run() NOT overridden; no direct __private calls; NFR patterns in code	Deduct 0.25 if run() overridden; 0.15 per missing required override
'a2c_inherited'	Extends SDLCAssistantAgent; _apply_policy() injects team-specific NFRs without removing A2C mandatory ones; DevRequest fields appropriate to domain	Same as e2a_inherited

9.3 Critic Response Format and Score Routing

```
# GeneratorCriticAgent is prompted to return ONLY valid JSON:
# {"score": 0.85, "passed": ["inherits BaseAgent", "run() not overridden"],
#  "missing": ["_fallback not overridden"], "regenerate": false}

# Score routing in generate() Step 9-10:
critic = GeneratorCriticAgent(config={'critic_type': critic_type,
                                      'model_id': config.get('critic_model_id',
                                          'anthropic.claude-3-haiku-20240307-v1:0')})
score, report = critic.score(response, request)
request['critic_report'] = report

if score >= 0.75:
    # Proceed to _write_output()
    pass
elif retry_count < 2:
    # Stricter regeneration
    response = self._fallback(request, config)
    retry_count += 1
else:
    return self._handle_error(
        NFRViolationError(f'Critic threshold 0.75 not met after 2 retries. '
                          f'Score: {score}. Missing: {report["missing"]}'),
        request, result, config
    )
```

10. GENERATORFACTORY AND DEVELOPERPLATFORMWORKFLOW

10.1 GeneratorFactory

Resolves the correct generator subclass from request['generator_type']. Follows the same approved-list governance pattern as E2A's agent routing.

```
class GeneratorFactory:
    _REGISTRY: Dict[str, type] = {
        'e2a_abstract':  E2AAbstractClassGenerator,
        'a2c_abstract':  A2CAbstractClassGenerator,
        'e2a_inherited': E2AInheritedClassGenerator,
        'a2c_inherited': A2CInheritedClassGenerator,
    }

    @classmethod
    def create(
        cls,
        request: GeneratorRequest,
        config: Dict[str, Any] = None
    ) -> BaseGeneratorAgent:
        gen_type = request.get('generator_type', '')
        cls_ = cls._REGISTRY.get(gen_type)
        if not cls_:
            raise ValueError(
                f'Unknown generator type: {gen type}. '
                f'Registered: {list(cls._REGISTRY.keys())}'
            )
        return cls_(config=config)

    @classmethod
    def register(cls, gen_type: str, generator_cls: type) -> None:
        """Register a new generator. Call at module import time."""
        cls._REGISTRY[gen_type] = generator_cls
```

10.2 DeveloperPlatformWorkflow

The top-level orchestrator. A developer passes a single GeneratorRequest and the workflow chains the correct generators automatically. The key chain: if generator_type='e2a_inherited' and abstract_class_source is not provided (defaults to 'generate'), it calls E2AAbstractClassGenerator first, injects the result as abstract_class_path, then calls E2AInheritedClassGenerator.

```
class DeveloperPlatformWorkflow:
    def __init__(self, config: Dict[str, Any] = None):
        self.config = config or {}

    def generate(
        self,
        request: GeneratorRequest,
        config: Dict[str, Any] = None
    ) -> GeneratorResult:
        config = config or self.config
        gen_type = request.get('generator_type', '')

        # Default: if abstract_class_source not provided, assume 'generate'
        if 'abstract_class_source' not in request:
            request['abstract_class_source'] = 'generate'

        # Chain: generate abstract class first if needed
        if (gen type in ('e2a inherited', 'a2c inherited')
                and request['abstract_class_source'] == 'generate'):
            abstract_gen_type = (
                'e2a_abstract' if gen_type == 'e2a_inherited' else 'a2c_abstract'
            )
            abstract_request = {
                **request.get('abstract_class_config', {}),
                'generator_type': abstract_gen_type,
                'runtime':        request.get('runtime', 'python'),
                'output_path':    request.get('output_path', './') + 'common/',
```

```
            }
            abstract_gen = GeneratorFactory.create(abstract_request, config)
            abstract_result = abstract_gen.generate(abstract_request, config)
            if not abstract_result['success']:
                return abstract_result  # propagate failure
            request['abstract_class_path']   = abstract_result['generated_files'][0]
            request['abstract_class_source'] = 'path'

        # Run main generator
        generator = GeneratorFactory.create(request, config)
        return generator.generate(request, config)
```

11. API EXPOSURE -- FASTAPI ROUTER

11.1 GeneratorAPIRouter

Each generator is exposed as a FastAPI POST endpoint under /api/v1/g2c/. The router is an E2A exposure layer -- no business logic, only parameter binding and response mapping. Public methods are exposed as APIs and can be linked to a UI-based application or an agentic AI-based orchestration system.

```
from fastapi import APIRouter, Depends
router = APIRouter(prefix='/api/v1/g2c', tags=['G2C Generator'])

@router.post('/generate', response_model=GeneratorResult)
async def generate(
    request: GeneratorRequest,
    config: dict = Depends(get_generator_config)
) -> GeneratorResult:
    '''
    Universal endpoint. Routes to correct generator via generator_type.
    generator_type: e2a_abstract | a2c_abstract | e2a_inherited | a2c_inherited
    '''
    return DeveloperPlatformWorkflow(config=config).generate(request, config)

@router.post('/e2a/abstract', response_model=GeneratorResult)
async def generate_e2a_abstract(
    request: GeneratorRequest,
    config: dict = Depends(get_generator_config)
):
    request['generator_type'] = 'e2a_abstract'
    return GeneratorFactory.create(request, config).generate(request, config)

@router.post('/a2c/abstract', response_model=GeneratorResult)
async def generate_a2c_abstract(
    request: GeneratorRequest,
    config: dict = Depends(get_generator_config)
):
    request['generator type'] = 'a2c abstract'
    return GeneratorFactory.create(request, config).generate(request, config)

@router.post('/e2a/inherited', response_model=GeneratorResult)
async def generate_e2a_inherited(
    request: GeneratorRequest,
    config: dict = Depends(get_generator_config)
):
    request['generator_type'] = 'e2a_inherited'
    return DeveloperPlatformWorkflow(config=config).generate(request, config)

@router.post('/a2c/inherited', response_model=GeneratorResult)
async def generate_a2c_inherited(
    request: GeneratorRequest,
    config: dict = Depends(get_generator_config)
):
    request['generator_type'] = 'a2c_inherited'
    return DeveloperPlatformWorkflow(config=config).generate(request, config)
```

11.2 JSON Configuration File -- generator-config.json

A developer can drive the full pipeline from a single JSON config file. The scaffold section (P0 config) is optional and only used by inherited class generators.

```
{
  "generator": {
    "source":                 "json",
    "generator_type":         "e2a_inherited",
    "runtime":                "python",
    "runtime_version":        "3.12",
    "platform":               "aws",
    "idempotency":            true,
    "async_mode":             true,
```

```
    "domain":                   "financial",
    "team":                     "settlement-team",
    "company":                  "acme-corp",
    "agent_name":               "SettlementAgent",
    "abstract_class_source":    "generate",
    "abstract_class_config":    { "base_class_name": "BaseSettlementAgent" },
    "user_prompt":              "SAP TM charge-to-settlement reconciliation agent",
    "system_prompt_additions": "Additional NFR: double-entry ledger invariant",
    "output_path":              "./generated/",
    "model_id":                 "anthropic.claude-3-5-sonnet-20241022-v2:0",
    "critic_model_id":          "anthropic.claude-3-haiku-20240307-v1:0"
  },
  "scaffold": {
    "build_tool":               "poetry",
    "project_name":             "SettlementEngine",
    "package name":             "settlement engine",
    "include_docker":           true,
    "include_makefile":         true,
    "dependencies":             ["fastapi>=0.111", "pydantic>=2.7", "langgraph>=0.1"]
  }
}
```

12. END-TO-END DEVELOPER FLOW

12.1 Full Stack Generation -- One Call

A developer needs a new E2A financial settlement agent from scratch. They provide one generator-config.json and call the API or workflow. The complete automated chain:

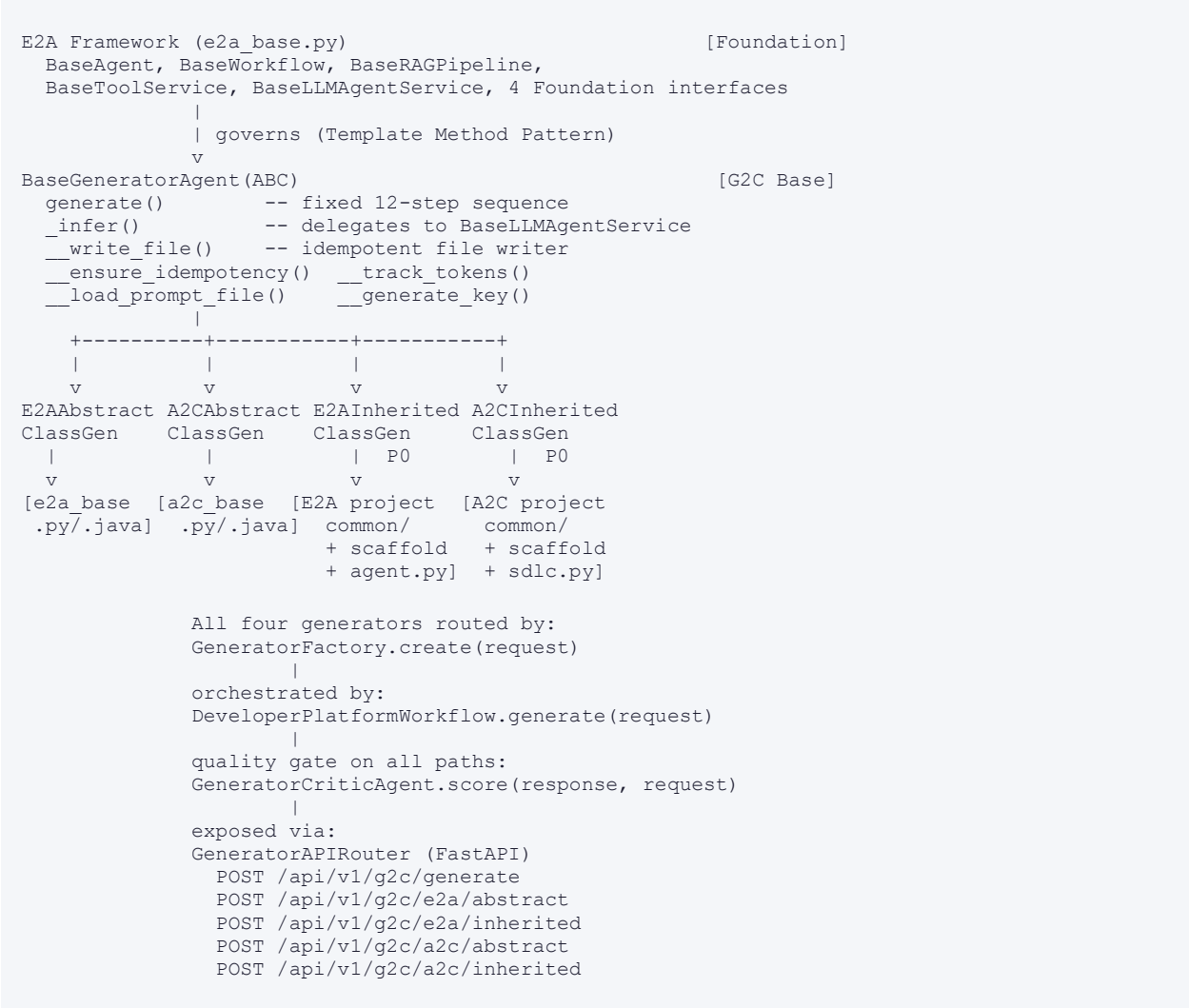
```
Developer provides generator-config.json
            |
            v
DeveloperPlatformWorkflow.generate(request, config)
  abstract_class_source defaults to 'generate' (not specified)
            |
            +-- Step A: E2AAbstractClassGenerator.generate()
            |     12-step sequence: LLM generates BaseSettlementAgent
            |     GeneratorCriticAgent: all 9 classes present, access modifiers correct
            |     Output: ./generated/common/e2a_base.py
            |     Injects abstract class path into main request
            |
            v
E2AInheritedClassGenerator.generate()
  Steps 1-4:  validate, governance, idempotency
  Steps 5-6:  load e2a_base.py from common/
  Steps 7-8:  P0 bootstrap -> pyproject.toml, dirs, Dockerfile, Makefile
              + copies e2a_base.py to project/common/
  Steps 9-11: LLM generates SettlementAgent(BaseSettlementAgent)
  Step  12:   GeneratorCriticAgent: inherits base, methods overridden, run() intact
  Steps 13-14: Write SettlementAgent.py + observability log
            |
            v
GeneratorResult {
  success: True,
  generated_files: [
    './generated/common/e2a_base.py',
    './generated/SettlementEngine/pyproject.toml',
    './generated/SettlementEngine/Dockerfile',
    './generated/SettlementEngine/Makefile',
    './generated/SettlementEngine/src/settlement_engine/agents/settlement_agent.py'
  ],
  critic_report: {score: 0.92, passed: [...], missing: []}
}
```

12.2 Common Developer Scenarios

Scenario	generator_type	abstract_class_source	What G2C produces
New E2A app, no base class exists	'e2a_inherited'	'generate' (default)	E2A abstract class + P0 scaffold + inherited agent class
New E2A app, team has existing base class	'e2a_inherited'	'path'	P0 scaffold + inherited agent class (base used as LLM context)
New E2A app, base class as pip package	'e2a_inherited'	'package'	P0 scaffold + inherited agent class (correct package imports)
Just the abstract base class file	'e2a_abstract'	N/A	Single e2a_base.py / E2ABase.java -- no scaffold
A2C workflow with custom SDLC logic	'a2c_inherited'	'generate'	E2A + A2C abstract files + P0 scaffold + SDLCAssistantAgent subclass

13. COMPLETE CLASS RELATIONSHIP MAP

How all G2C classes relate to each other and to the E2A, A2C, and P0 frameworks:



14. REPOSITORY STRUCTURE AND PROMPT FILE LAYOUT

```
a2c-framework/                              # G2C files live alongside A2C
  # ── G2C Core ──────────────────────────────────────────────────────────
  base_generator_agent.py                    BaseGeneratorAgent ABC
  e2a_abstract_class_generator.py            E2AAbstractClassGenerator
  a2c_abstract_class_generator.py            A2CAbstractClassGenerator
  e2a_inherited_class_generator.py           E2AInheritedClassGenerator
  a2c_inherited_class_generator.py           A2CInheritedClassGenerator
  generator_critic_agent.py                 GeneratorCriticAgent
  generator_factory.py                      GeneratorFactory
  developer_platform_workflow.py             DeveloperPlatformWorkflow
  generator_api_router.py                   FastAPI endpoints
  typedicts.py                              GeneratorRequest, GeneratorResult

  # ── Versioned System Prompts ──────────────────────────────────────────
  prompts/
    e2aabstractclassgenerator_system_v1.txt
    a2cabstractclassgenerator_system_v1.txt
    e2ainheritedclassgenerator_system_v1.txt
    a2cinheritedclassgenerator_system_v1.txt
    generatorcriticagent system v1.txt

  # ── JSON Schemas ──────────────────────────────────────────────────────
  schemas/
    generator-config-schema.json            JSON Schema for generator-config.json
    scaffold-config-schema.json             From P0 Framework (reused here)

  # ── Examples ──────────────────────────────────────────────────────────
  examples/
    e2a abstract config.json                Minimal: generate E2A base class only
    e2a_inherited_minimal_config.json       Minimal: e2a_inherited with all defaults
    e2a_inherited_full_config.json          Full: all options explicit
    a2c_abstract_config.json                Generate A2C base class only
    a2c_inherited_config.json               Full A2C inherited class pipeline
    java spring boot config.json            Java runtime example
```

15. ENVIRONMENT VARIABLE REFERENCE

Central configuration: every key in this document is defined once, in the E2A_CONFIG registry at the top of e2a_base.py (Implementation Playbook Section 5). Resolution order is call-time kwargs > config dict > environment variable > registered default, performed only by resolve_config(). Projects add their own keys with register_config_key() in src/settings.py. .env.example and the Playbook configuration table are generated from the registry (env_example(), config_reference_markdown()), so they cannot drift from the code. The generator variables below (GENERATOR_*, CRITIC_MODEL_ID) are registered through register_config_key() in the generator's settings module alongside the framework keys, and the projects G2C generates inherit the same registry rather than defining their own environment handling.

Variable	Used in	Default	Controls
LLM_MODEL_ID	BaseLLMAgentService (model_id, via config["llm_service"])	'anthropic.claude-3-5-sonnet-20241022-v2:0'	Main LLM for all generation calls
LLM_POD_CONCURRENCY / LLM_MAX_RPM / LLM_MAX_TPM / LLM_REPLICA_COUNT	BaseLLMAgentService	32 / 600 / 300000 / 1	Capacity governance of generation calls; full key list in the Implementation Playbook, Section 5
CRITIC_MODEL_ID	GeneratorCriticAgent	'anthropic.claude-3-haiku-20240307-v1:0'	Cheaper model for quality gate (FinOps)
GENERATOR_MAX_LATENCY_MS	generate() Step 12	60000	Latency SLO -- code generation takes longer than agent calls
GENERATOR_MAX_TOKENS	generate() Step 7	32000	Token budget for class generation
GENERATOR_MIN_CRITIC_SCORE	generate() Step 9	0.75	Quality gate threshold -- regenerate if below
GENERATOR_PROMPT_DIR	BaseGeneratorAgent.__load_prompt_file()	'prompts/'	Base directory for versioned system prompt files
BOOTSTRAP_OUTPUT_PATH	P0 BaseProjectBootstrapper	'./'	Default output directory for P0 scaffold
GENERATOR_IDEMPOTENCY_STORE	BaseGeneratorAgent.__ensure_idempotency()	'local'	'local' checks directory; 'dynamodb' for distributed

16. IMPLEMENTATION CHECKLIST

New Generator Subclass

- Never call a model SDK from a generator — model calls go through _infer() to the bound config['llm_service']
- Define generator_name class attribute -- governance key, used in all log entries
- Add generator_type to GeneratorFactory._REGISTRY
- Override _validate_request(): check all required GeneratorRequest fields for this generator
- Override _resolve_config(): handle source='json' and source='prompt'
- Override _build_system_prompt(): call __load_prompt_file() + inject runtime patterns + merge additions
- Override _build_user_prompt(): construct developer-facing prompt from request fields
- Override _evaluate_output(): call GeneratorCriticAgent with correct critic_type
- Override _write_output(): write generated files with runtime-correct filename, return paths
- Override _fallback(): return stricter regeneration prompt string
- Override _handle_error(): log error, append to errors[], return safe GeneratorResult
- DO NOT override generate() -- the 12-step sequence is enforced by the base class
- DO NOT call any __private method from outside the class
- Set max_latency = 60,000 ms (LLM code generation takes longer than agent calls)
- Set max_tokens = 32,000 (class generation requires large token budget)

New Prompt File

- Name: prompts/{generator_class_name_lowercase}_system_v{N}.txt
- Include: Template Method Pattern rules, access modifier specification, mandatory NFRs, runtime pattern placeholders ({runtime_patterns})
- Use {placeholders} for runtime, runtime_version, idempotency, async_mode, domain, system_prompt_additions
- Test: verify generated output passes GeneratorCriticAgent at score >= 0.75 before committing
- Version: increment N only for breaking changes; old versions remain available for rollback

JSON Config File Validation

- Validate against schemas/generator-config-schema.json before passing to generate()
- generator section required; scaffold section optional (only for inherited generators)
- abstract_class_source defaults to 'generate' if omitted -- no explicit field required
- system_prompt_additions defaults to empty string -- base system prompt always valid without it